

Opportunistic Knowledge Adaption in Self-Learning Systems

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Abstract—In the context of Autonomous Learning, the question arises how an online learning system adapts its knowledge according to a changing environment, i.e. arrival of new classes or changing noise functions, to maintain a robust level of performance. As a solution, we suggest an architectural design inspired by a variant of the Observer/Controller framework. We present a scenario, in which the presented architecture is assumed to improve the performance, because the system is aware of currently available knowledge and can opportunistically exploit this knowledge.

Keywords—Organic Computing, Online learning, Autonomous Learning, Knowledge Adaption.

1. Motivation and challenges

In the year 1993, Marc Weiser postulated his vision about a future in which the number of devices and their degree of connectivity explodes [1]. He claimed that the interfaces should vanish unobtrusively into background. Later in 2003, the Autonomic Computing initiative [2], [3] and 2004 the Organic Computing initiative [4], [5] described blueprints for systems that are aware of themselves and their environment.

Driven by these general concepts, the last two decades have witnessed a strong trend towards autonomous systems, i.e. a shift of responsibilities from engineers to the systems themselves. As a reaction to the need of finding appropriate reactions to unanticipated conditions, changing system constellations, and disturbances or even attacks, the technical systems shall be increasingly equipped with capabilities to self-adapt their behaviour, to identify new strategies, and to alter their integration status in the overall system structure. Such a process requires some kind of creativity – which can be realised by machine learning techniques.

The basis for these development lies in the sub-domain of machine learning that focuses on learning at runtime without the need of, e.g. large amounts of training data is called 'autonomous learning'. By now, most of the self-adaptive and self-organising systems (SASO) have made use of a dedicated and problem-based usage of specific machine learning techniques [6], [7]. However, the basic idea of this PhD project is that this is not sufficient for dynamically changing problems, open environments and hidden mutual influences among the behaviour of autonomous subsystems.

Consequently, the main challenge of this PhD project is the following:

“How can we improve the utilisation of different dynamically available knowledge sources as basis for learning the appropriate self-adaption behaviour in SASO systems based on experiences?”

We call this concept 'opportunistic meta-learning', since it tries to make the best of the currently available resources for improving the run-time learning behaviour at a meta level [8]. In the remainder of this abstract we present the contribution and objectives (Section 2), the methodology and preliminary results (Section 3), and finally, the future work and a research plan (Section 4) of the PhD project.

2. Contribution and objectives

In general, we develop approaches for a continuous adaption of the knowledge of an intelligent system in an opportunistic way, which means exploiting available knowledge at run-time. The latter requires an appropriate model to capture the state of knowledge sources in order to rank the knowledge sources according to an intelligent system's current needs. Knowing which knowledge sources have to be used and given a temporarily available pool of such instances, the question comes up according to which protocol the knowledge transfer could be organised. Finally, in order to manage them correctly, their status should be monitored over time, and there should be developed an update routine that is activating or deactivating knowledge sources. In order to solve this, the following research objectives can be summarised:

- How to define a model of knowledge sources that enables learning systems to take advantage of spontaneously appearing knowledge?
- Which protocol should be used to enable communication between the intelligent system and the knowledge source?
- How to manage knowledge sources by an intelligent system?
- How to activate or deactivate available knowledge sources?

3. Methodology and preliminary results

In this section, we suggest an opportunistic learning architecture that we used for the domain of quality assessment

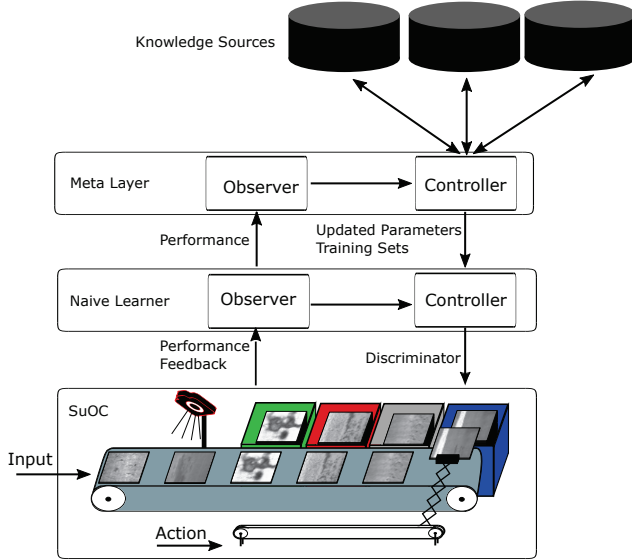


Figure 1. Quality assessment as a scenario for opportunistic self-awareness of meta-learning components.

(QA).

In the traditional Observer/Controller framework, there are only two types of layers. Layer 0 is called System under Control and Observation (SuOC) and the upper layers are called Control Mechanisms (CM) [9]. Introducing the concept of knowledge adaption we require an active system, a learning instance, and the knowledge adaption. Hence, we stack up three layers upon each other as seen in the Figure 1, namely, the SuOC at the bottom, the naïve learner in the middle, and the meta layer on top. The layers of the knowledge adaption framework are created bottom-up but monitoring and controlling is done top-down. That means, the layers do not rely on their upper layers but the upper layers contribute to the system as a whole. Instead of using a bottom-up escalation mechanism, which would add another responsibility to the layer, the observer in the Observer/Controller control mechanism one layer above monitors the behaviour of the layer. This results in a stack of systems that are only monitoring downwards and are still capable of working without their managing superior layers.

As a first case study, we use the domain of QA, where the task is to analyse the quality of produced workpieces to decide whether the products can be sold or not. This includes different error classes besides a defect-free class for different quality levels. In our approach, the basic control loop is realised by means of observation, control, and productive parts (i.e., the surface inspection system as a SuOC).

The SuOC in Figure 1 is assigning the workpieces to different buckets (for delivery or failure classes), while the decision is guided by the adaption loop. The system starts without knowledge and has to make use of the dynamic sources (such as humans, best practises, or other systems) depending on their availability, certainty, or cost. In the simulation of the conveyor belt, we are streaming images of

a hot-rolled steel strip to the SuOC. We use the Northeastern University (NEU) surface defect data set [10] with 6 error types providing 300 images each. After receiving the sample, the SuOC classifies it according to a static discriminator. The classification subsequently trigger a feedback responding with the ground truth label with a certain probability.

These information is dynamically observed by the naïve learner above which compares the predictions with the truth values and calculates a performance value. In our application, we used a history-based accuracy. When the performance falls below a certain threshold, the samples and truth labels are used to train a static machine learning model. After training, the old discriminator is replaced by the new model if its performance is better.

On a higher level, the meta layer extends the observed knowledge with knowledge from the top k knowledge sources ranked by an knowledge value that is based on the information gain of their sample and class distribution [11], the probability of the knowledge source classes to appear in the input steam, and being misclassified by the naïve learner. Furthermore, we represent knowledge sources containing classified images of the NEU surface defect data set which are not used for the streaming. The knowledge sources lack in quality containing unseen classes, noise, or unreliable class assignments. Finally, the performance in terms of accuracy is monitored while stressing the system with noisy images, new classes, or unreliable knowledge sources.

4. Future work and research plan

The paper has shown the concepts of a PhD project in the domain of self-adaptive systems. After presenting the scope of the work, a concrete O/C architecture with a specification is presented which is demonstrated in the domain of quality assessment. Given the presented scenario in quality assessment, a possible layer on top of the O/C layers is imaginable which aggregates the whole knowledge learned so far and is able to exchange this knowledge package and ask for more packages from other learning systems that provide the same communication layer. Besides the simulation of the conveyor belt, other scenarios are planned to be investigated. For example, finding the shortest path to a target within a randomly generated maze is worth to be analysed because of the limited sensor view and the local restrictions of agents distributed at random positions in the maze. As the imperfect knowledge is another difficulty in this scenario, the agents might then communicate over a channel in order to help each other finding the target as fast as possible. As random maze generation is computationally cheap, the usage of knowledge of previous runs as starting point can be easily evaluated. Other scenarios are planned to show certain aspects of the knowledge-self adaptive system. We think that the future of learning especially in the domain of predictive maintenance cannot be handled with static machine learning models engineered at design-time as there are always unpredictable observations to which the system should react to at run-time.

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